

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Data Mining (DM)

Fall 2025

Lecture 6b: Know your data



**DATA, DATA; WHERE
DO YOU LIE?**



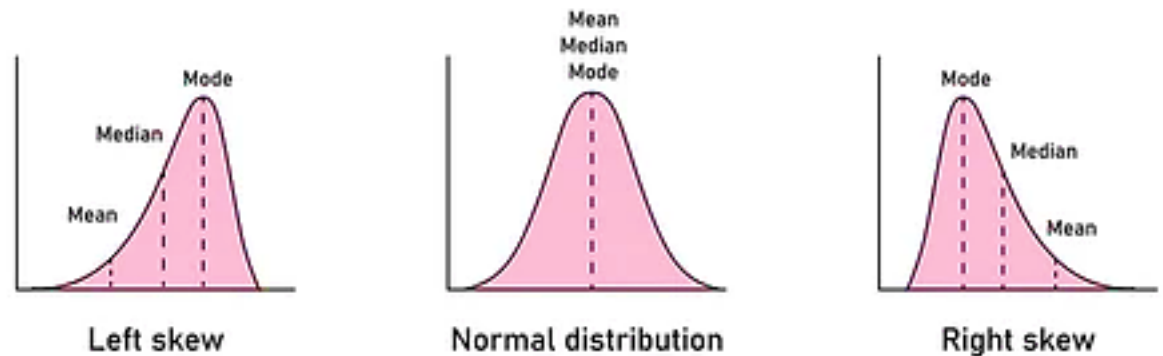
Data, Data; Where do You Lie?

- I Lie in the Center
- I Lie in the Dispersion

Central Tendency of Data

- Mean
- Median
- Mode

Mean, Median and Mode



Mean

- Let $x_1, x_2, \dots, x_N$ be a set of N values or *observations*, such as for some numeric attribute X , like salary
- The **mean** of this set of values is

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \dots + x_N}{N}. \quad (2.1)$$

Example 2.6 Mean. Suppose we have the following values for *salary* (in thousands of dollars), shown in increasing order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110. Using Eq. (2.1), we have

$$\begin{aligned}\bar{x} &= \frac{30 + 36 + 47 + 50 + 52 + 52 + 56 + 60 + 63 + 70 + 70 + 110}{12} \\ &= \frac{696}{12} = 58.\end{aligned}$$

Thus, the mean salary is \$58,000. ■

Weighted Arithmetic Mean

- Sometimes, each value x_i in a set may be associated with a weight w_i
 - The weights reflect the significance, importance, or occurrence frequency attached to their respective values

$$\bar{x} = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i} = \frac{w_1 x_1 + w_2 x_2 + \cdots + w_N x_N}{w_1 + w_2 + \cdots + w_N}. \quad (2.2)$$

Weigthed Mean Example

- Quiz Average?

Quiz	Marks out of 10
Q1	2
Q2	4
Q3	3
Q4	7

$$\text{Average} = (2+4+3+7)/4 = 16/4 = 4$$

Weighted Mean Example

- Quiz Average?

Quiz	Marks out of 10	Weight
Q1	2	1
Q2	4	1
Q3	3	1
Q4	7	2

$$\begin{aligned}\text{Weighted Mean} &= (2 \times 1) + (4 \times 1) + (3 \times 1) + (7 \times 2) \\ &= (2 + 4 + 3 + 14) / 5 = 23 / 5 = 4.6\end{aligned}$$

Mean

- A useful Measure of Central Tendency of Data
- But has some Issues
 - sensitivity to extreme values
 - Even a small number of extreme values can corrupt the mean
 - The mean salary at a company may be substantially pushed up by that of a few highly paid managers

Trimmed Mean

- Useful in avoiding Sensitivity to extreme values
- Mean obtained after chopping off values at the high and low extremes
 - For example, sort the values observed for *salary* and remove the top and bottom 2% before computing the mean
 - Avoid trimming too large a portion (such as 20%) at both ends

Mean

- A useful Measure of Central Tendency of Data **when data is Symmetric**
- **Not** a good measure for Asymmetric (Skewed) Data

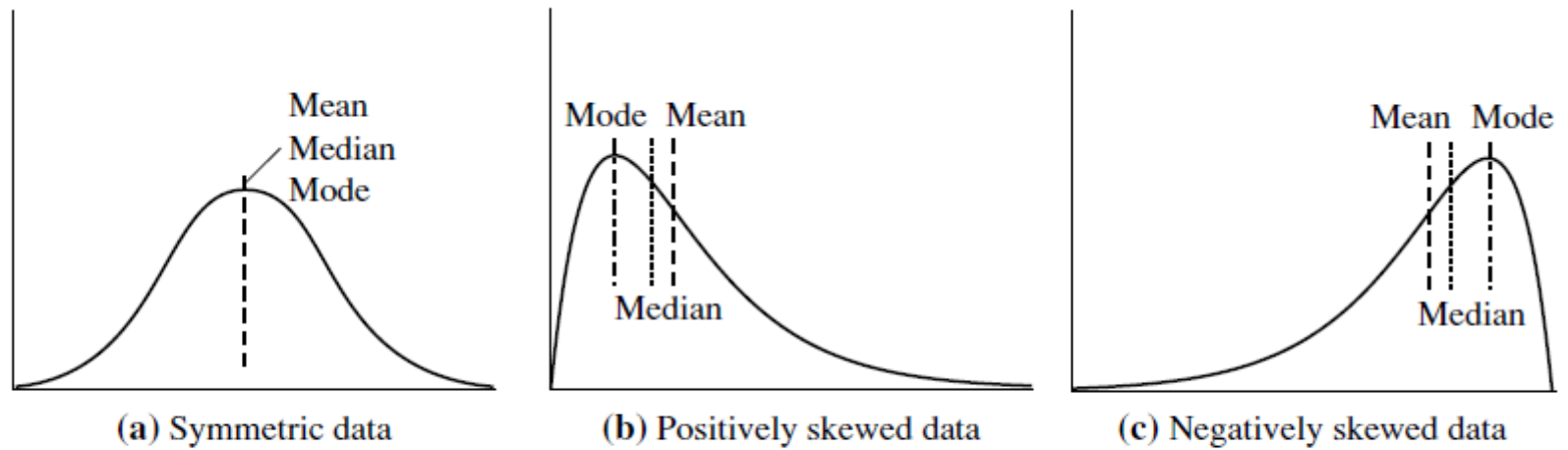


Figure 2.1 Mean, median, and mode of symmetric versus positively and negatively skewed data.

Median

- A ***better measure*** of the center of data, for *skewed* (asymmetric) data
- It is the *middle value* in a set of ordered (sorted) data values
 - i.e. it is the value that separates the higher half of a data set from the lower half.
- We may extend the concept to **ordinal data**

Example 2.7 Median. Let's find the median of the data from Example 2.6. The data are already sorted in increasing order. There is an even number of observations (i.e., 12); therefore, the median is not unique. It can be any value within the two middlemost values of 52 and 56 (that is, within the sixth and seventh values in the list). By convention, we assign the average of the two middlemost values as the median; that is, $\frac{52+56}{2} = \frac{108}{2} = 54$. Thus, the median is \$54,000.

Suppose that we had only the first 11 values in the list. Given an odd number of values, the median is the middlemost value. This is the sixth value in this list, which has a value of \$52,000. ■

Example 2.6 Data: 30, 36, 47, 50, 52, **52, 56**, 60, 63, 70, 70, 110

Median

- Expensive to compute when we have a large number of observations
- Easily *approximate* the value
 - For numeric attributes only

$$median = L_1 + \left(\frac{N/2 - (\sum freq)_l}{freq_{median}} \right) width, \quad (2.3)$$

Median

- L_l
 - lower boundary of the **median interval**
- N
 - number of values in the entire data set

$$median = L_l + \left(\frac{N/2 - (\sum freq)_l}{freq_{median}} \right) width, \quad (2.3)$$

Median

- $(\sum freq)_l$
 - sum of the frequencies of all of the intervals that are lower than the median interval
- $freq_{median}$
 - frequency of the median interval, and
- $width$
 - width of the median interval

$$median = L_1 + \left(\frac{N/2 - (\sum freq)_l}{freq_{median}} \right) width, \quad (2.3)$$

Mode

- Value that occurs most frequently
- Can be determined for qualitative and quantitative attributes
- Unimodal, Bimodal, and Trimodal data

Mode

- Unimodal Data
 - One value occurs most frequently
- Bimodal Data
 - Two values occur most frequently
 - (with same frequency)
- Trimodal Data
 - Three values occur most frequently
 - (with same frequency)

Example 2.8 Mode. The data from Example 2.6 are bimodal. The two modes are \$52,000 and \$70,000. ■

Example 2.6 Data: 30, 36, 47, 50, **52, 52**, 56, 60, 63, **70, 70**, 110

Mode

- In general, a data set with two or more modes is multimodal
- At the other extreme, if each data value occurs only once, then there is no mode

Midrange

- The midrange can also be used to assess the central tendency of a numeric data set
- It is the average of the largest and smallest values in the set

Example 2.9 Midrange. The midrange of the data of Example 2.6 is $\frac{30,000+110,000}{2} = \$70,000$. ■

Example 2.6 Data: **30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110**



Dispersion of Data

- Range, Quartiles, and Interquartile Range
- Variance and Standard Deviation

Range

- The range of a data set is the difference between the largest and smallest values

Example 2.6 Data: **30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110**

Range: $110 - 30 = 80$

Why Dispersion of Data?

- 1, 3, 5 (Mean = 3), Dispersion/spread? 4
 - 2, 3, 4 (Mean = 3) 2
 - 3, 3, 3 (Mean = 3) 0
-
- Range, Quartiles, and Interquartile Range
 - Variance and Standard Deviation

Choose the mobile, M1 or M2?

- M1
 - Customer ratings: 1, 1, 1, 1, 5, 5, 5, 5
 - Mean = 3, Dispersion = $5 - 1 = 4$
- M2
 - Customer ratings: 2, 3, 3, 3, 3, 3, 3, 4
 - Mean = 3, Dispersion = $4 - 2 = 2$
- Which mobile you will prefer to buy?
 - M1 or M2

Measuring the Dispersion of Data

- Quartiles, outliers and boxplots

- **Quartiles:** Q_1 (25th percentile), Q_3 (75th percentile)

- **Inter-quartile range:** $IQR = Q_3 - Q_1$

- **Five number summary:** min, Q_1 , median, Q_3 , max

- **Boxplot:** ends of the box are the quartiles; median is marked; add whiskers, and plot outliers individually

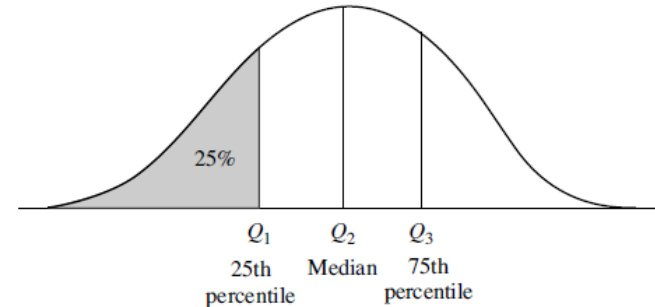
- **Outlier:** usually, a value higher/lower than $1.5 \times IQR$ from Q_3/Q_1

- Variance and standard deviation

- **Variance:** (algebraic, scalable computation)

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^n (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^n x_i^2 - \mu^2$$

- **Standard deviation** s (or σ) is the square root of variance s^2 (or σ^2)



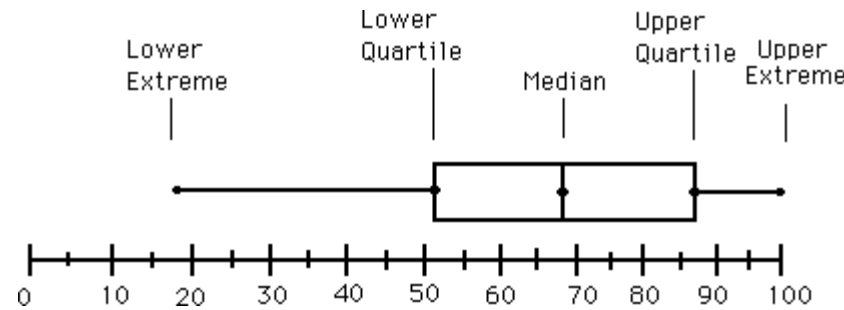
Example: Dispersion of Data

- 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110
- Q1
- Q2
- Q3
- IQR
- Five Number Summary
- Variance
- SD

Graphic Displays of Basic Statistical Descriptions

- **Boxplot:** graphic display of five-number summary
- **Histogram:** x-axis are values, y-axis repres. frequencies
- **Quantile plot:** each value x_i is paired with f_i indicating that approximately $100 f_i \%$ of data are $\leq x_i$
- **Quantile-quantile (q-q) plot:** graphs the quantiles of one univariant distribution against the corresponding quantiles of another
- **Scatter plot:** each pair of values is a pair of coordinates and plotted as points in the plane

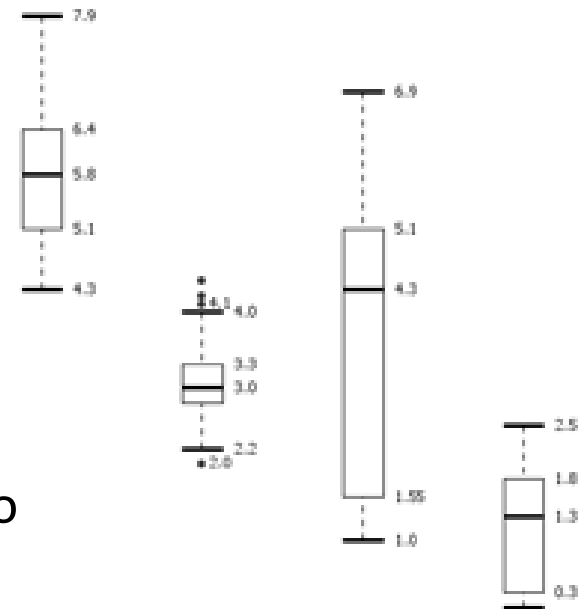
Boxplot Analysis



- **Five-number summary** of a distribution
 - Minimum, Q1, Median, Q3, Maximum

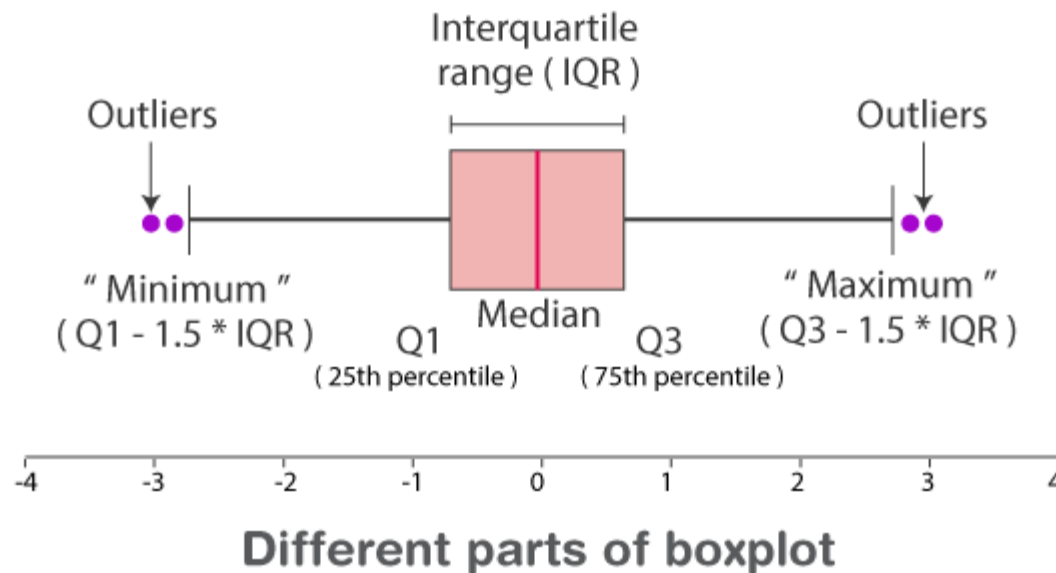
- **Boxplot**

- Data is represented with a box
- The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
- The median is marked by a line within the box
- Whiskers: two lines outside the box extended to Minimum and Maximum
- Outliers: points beyond a specified outlier threshold, plotted individually



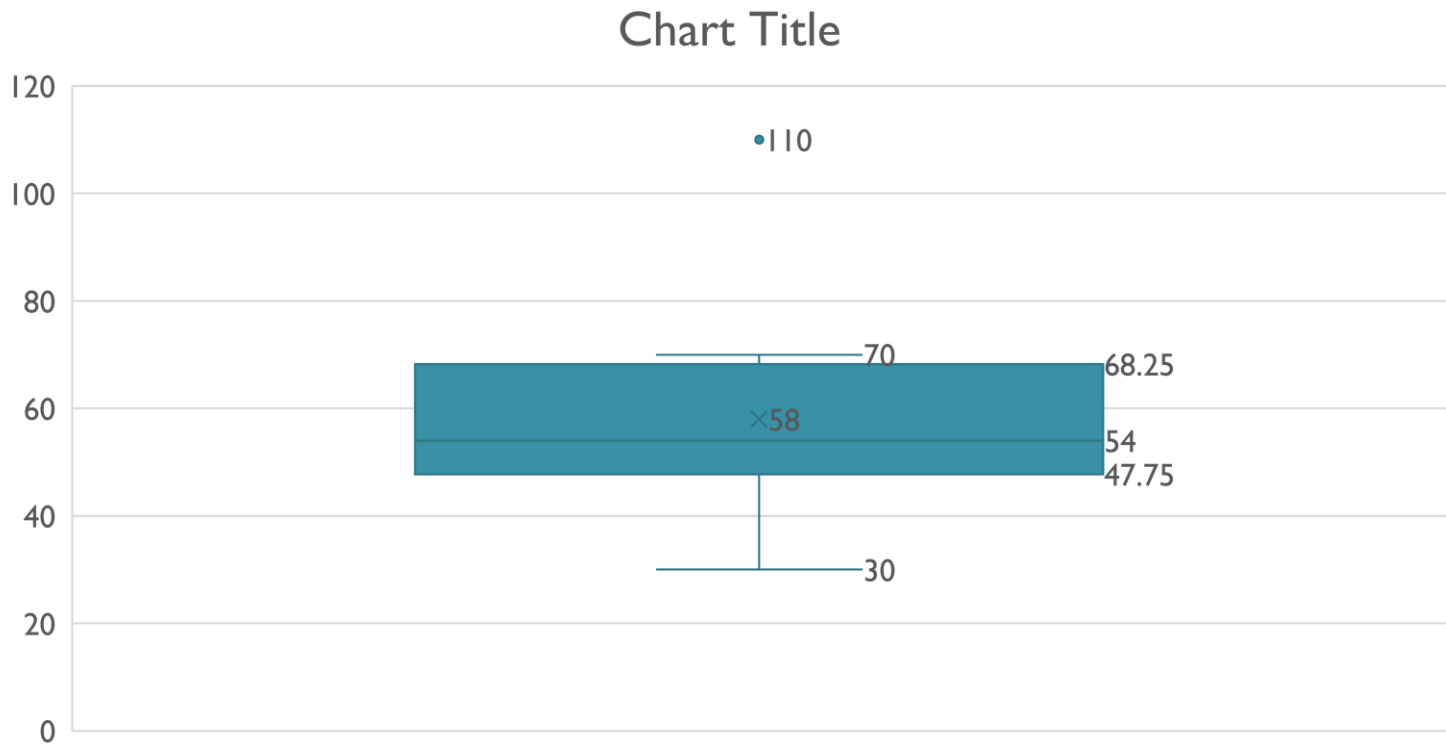
Boxplot in Matlab

- `>> d = [30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110];`
- `>> boxplot(d);`



Boxplot in Matlab

- `>> d = [30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110];`
- `>> boxplot(d);`

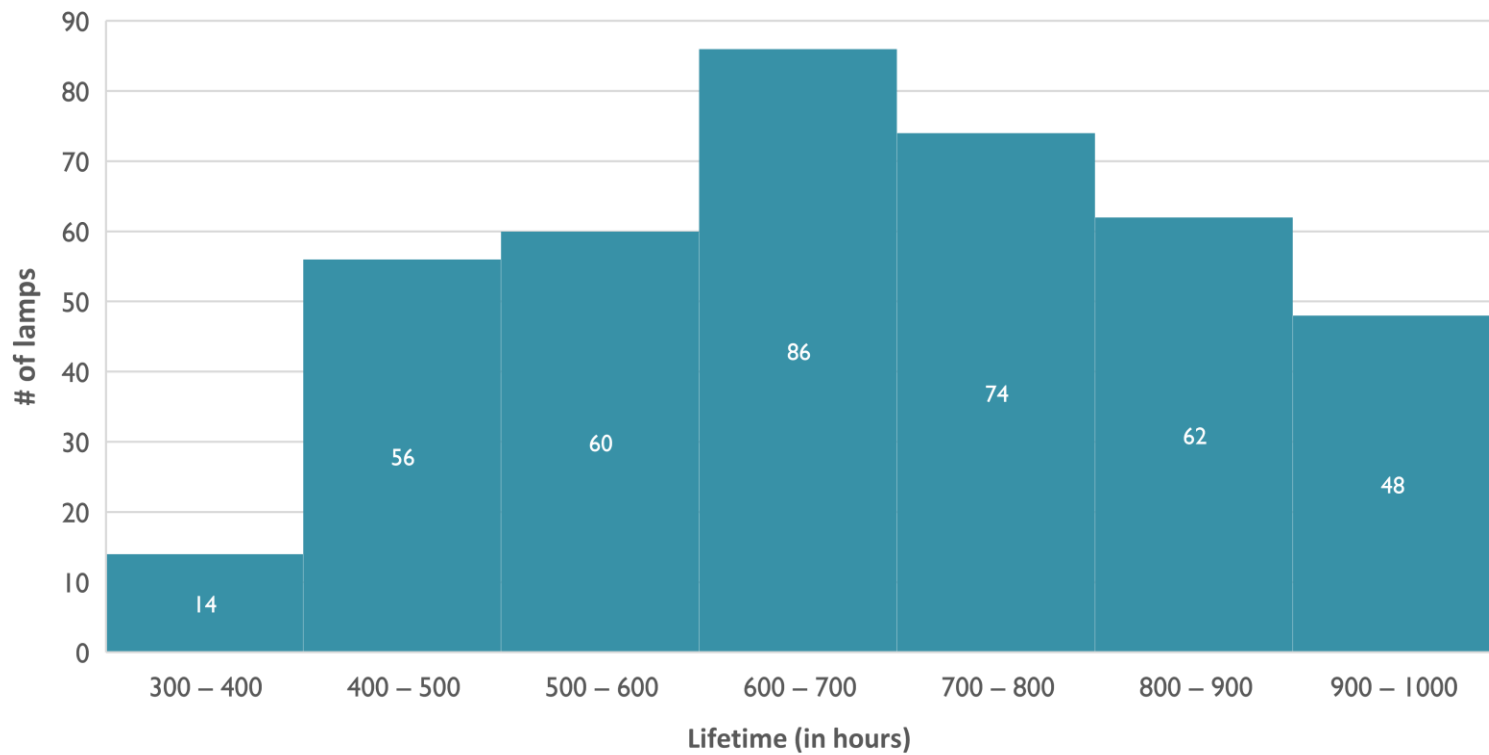


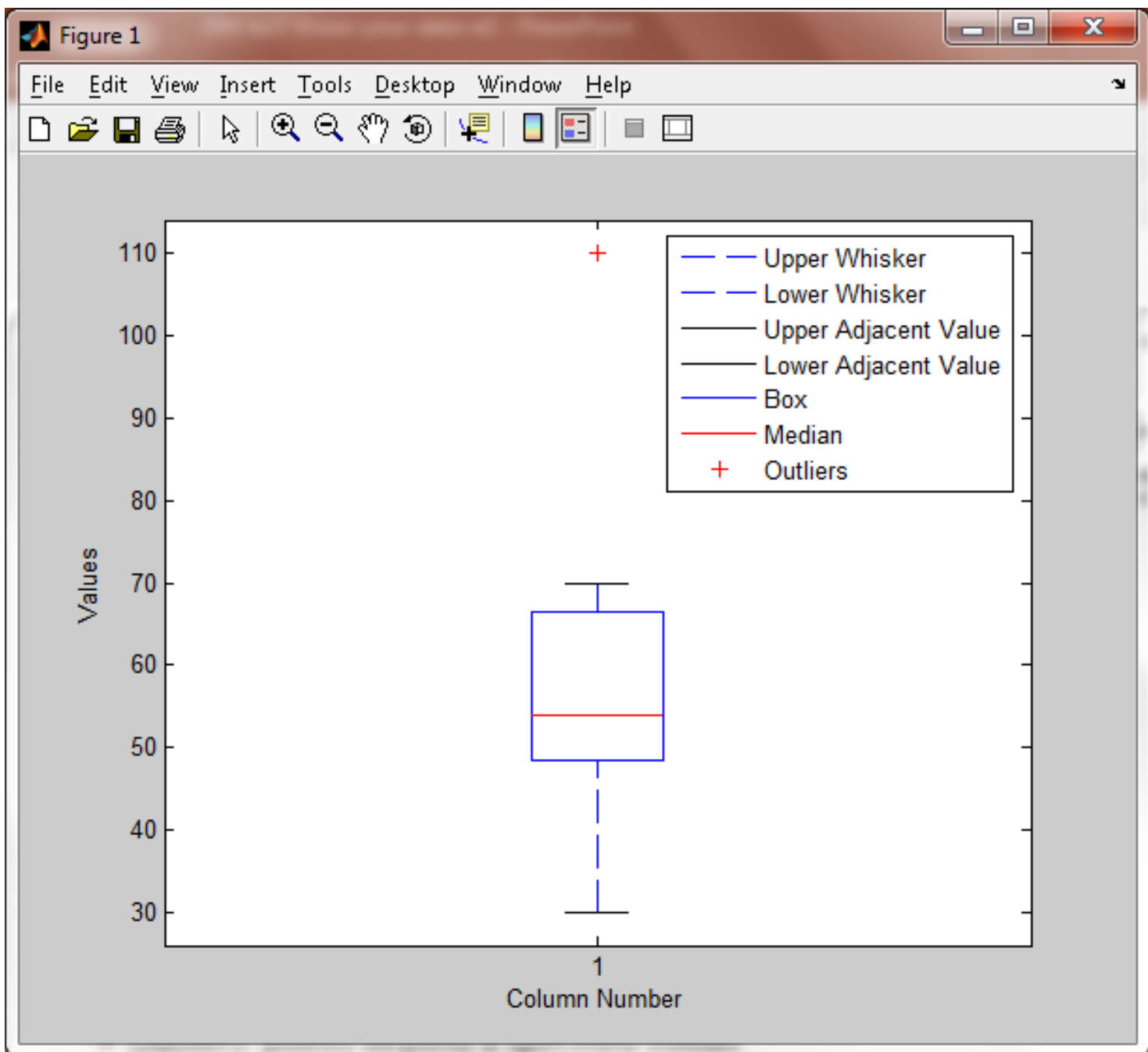
Histogram Example

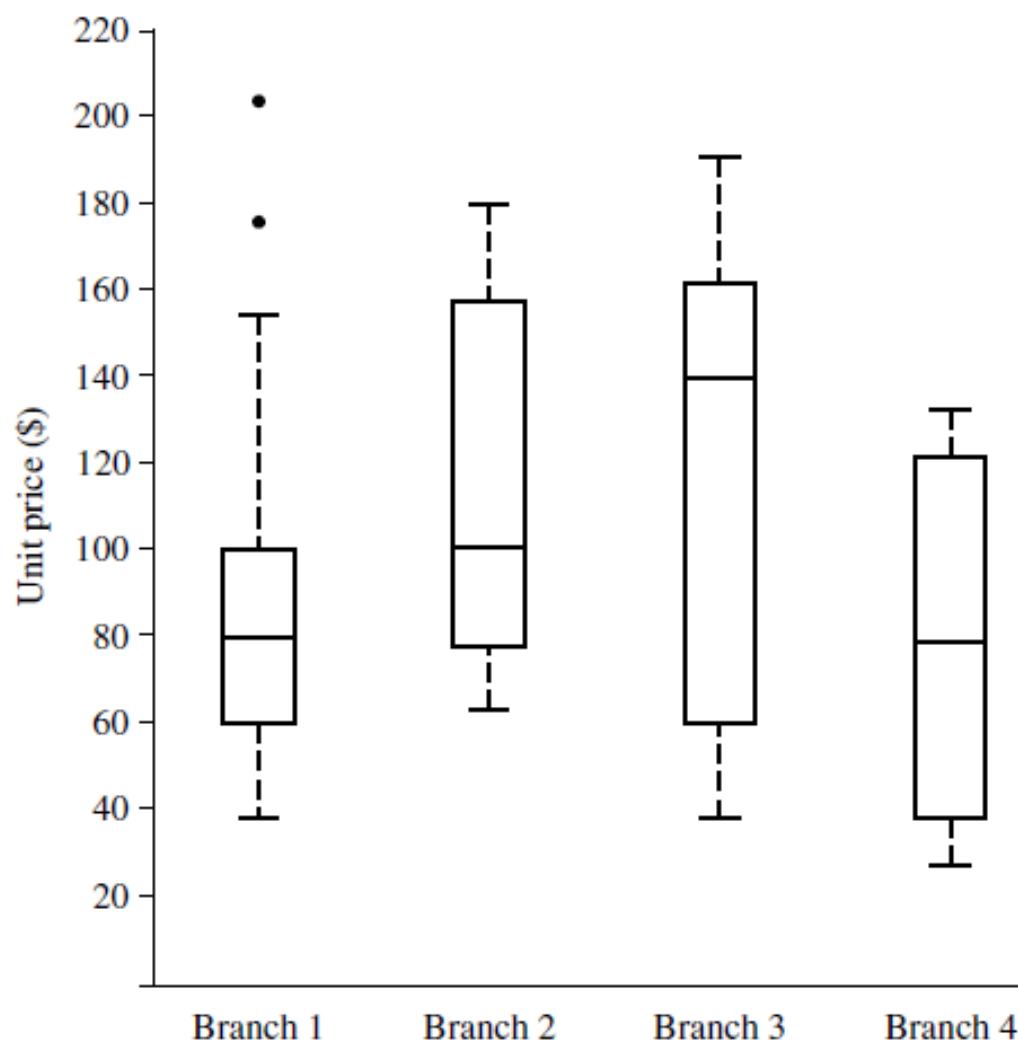
Question: The following table gives the lifetime of 400 neon lamps. Draw the histogram for the below data.

Lifetime (in hours)	Number of lamps
300 – 400	14
400 – 500	56
500 – 600	60
600 – 700	86
700 – 800	74
800 – 900	62
900 – 1000	48

Histogram Example



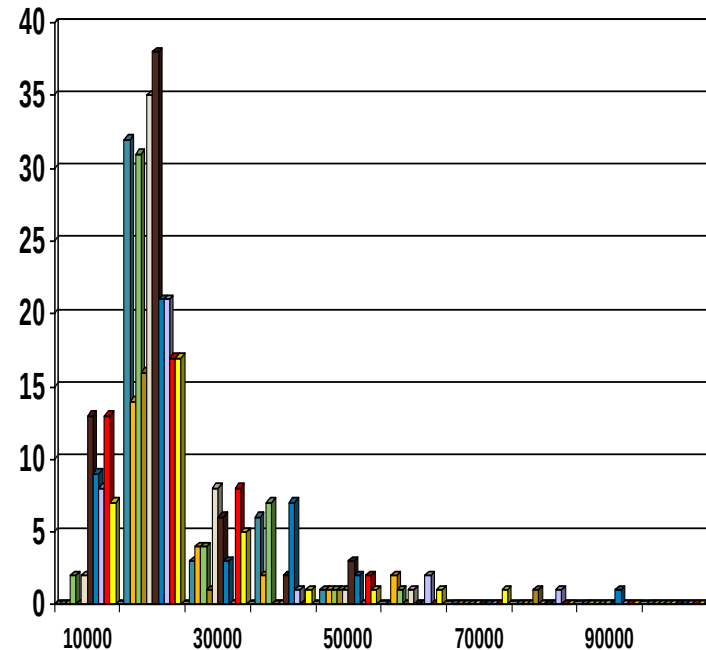




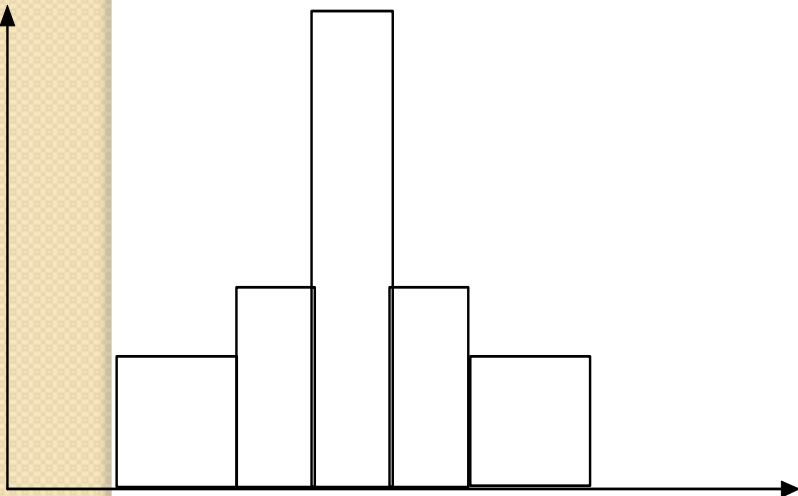
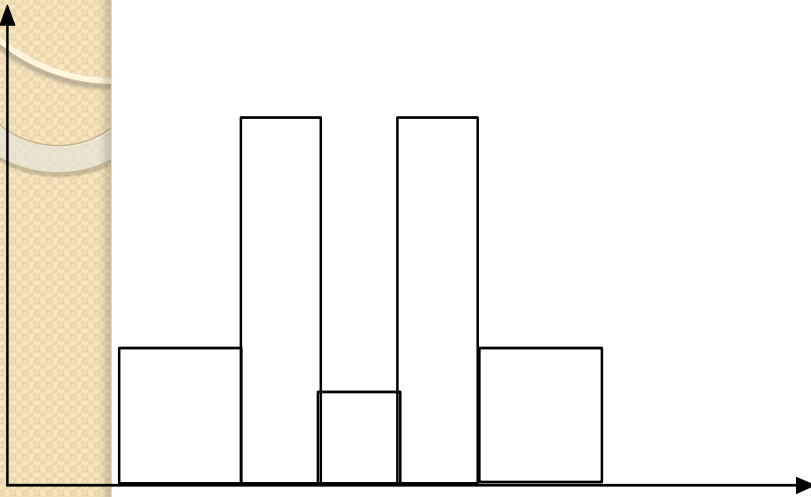
Boxplot for the unit price data for items sold at four branches of *AllElectronics* during a given time period.

Histogram Analysis

- Histogram: Graph display of tabulated frequencies, shown as bars
- It shows what proportion of cases fall into each of several categories
- The categories are usually specified as non-overlapping intervals of some variable. The categories (bars) must be adjacent



Histograms Often Tell More than Boxplots



- The two histograms shown in the left may have the same boxplot representation
 - The same values for: min, Q1, median, Q3, max
- But they have rather different data distributions

Properties of Normal Distribution Curve

- The normal (distribution) curve
 - From $\mu - \sigma$ to $\mu + \sigma$: contains about 68% of the measurements (μ : mean, σ : standard deviation)
 - From $\mu - 2\sigma$ to $\mu + 2\sigma$: contains about 95% of it
 - From $\mu - 3\sigma$ to $\mu + 3\sigma$: contains about 99.7% of it

